

Onyekachi Emenike, PhD

Data Scientist | PhD Computational Mathematics

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PROFILE

Data scientist with a PhD in Computational Mathematics and five peer-reviewed publications on quasi-Monte Carlo methods for high-dimensional problems. I build machine learning models end to end, from data collection, EDA and feature engineering to explainable, tracked models, and put them into production with orchestration, CI/CD and drift monitoring. Experienced with regression, time-series forecasting, clustering and PCA, and retrieval-augmented AI grounded in real sources. Former lecturer in Data Analytics & Machine Learning (MSc), used to explaining quantitative results to non-technical audiences. Based in Germany with EU work authorization.

SELECTED PROJECTS

House Price Predictor — Rhineland-Palatinate

Python · SQLite · XGBoost · SHAP · MLflow · Streamlit · Plotly

- Built an end-to-end ML pipeline for 3,800+ regional real estate listings: database schema design, full EDA, feature engineering and an XGBoost regressor reaching R^2 0.75 on held-out data.
- Tracked experiments in MLflow and shipped with SHAP explainability and an LLM chat assistant, so users see why a price was predicted, not just the number.

End-to-End MLOps Pipeline — Demand Forecasting

Python · MLflow · Prefect · Docker · Kubernetes · GitHub Actions · Evidently

- Engineered a pipeline from raw data ingestion through modelling to deployment, with Prefect orchestration, GitHub Actions CI/CD and Docker/Kubernetes deployment.
- Added automated data and model drift monitoring with Evidently, so degradation is flagged instead of relying on manual review.

Maternal Health AI Assistant — Agentic RAG

LangChain · OpenAI GPT-4o · FAISS · Hugging Face · Streamlit

- Built an agentic RAG system in which a LangChain agent answers from a FAISS-indexed store of WHO/AAP/CDC guidelines or calls the USDA FoodData Central API for live nutrient data.
- Deployed live on Streamlit; every answer is traceable to a published guideline or live data source rather than model memory.

Clustering & PCA for Equity Portfolio Structure

Python · scikit-learn · SciPy · pandas · Matplotlib

- Applied hierarchical clustering (correlation-based distance, linkage chosen by cophenetic correlation) and PCA to 20 years of daily returns for 50 U.S. large-cap stocks.
- Identified 5 coherent clusters and showed one market factor explains 41% of return variance, with ~20 components needed for 80%.

Deep Learning for Multi-Asset Return Forecasting

Python · TensorFlow/Keras · pandas · NumPy · statsmodels

- Built the data pipeline and EDA (ADF stationarity tests, correlation structure) for 5 ETFs over 15 years of daily data; designed a 3-layer stacked LSTM forecasting 25-day-ahead returns from multi-horizon lagged features.
- Authored the single- vs. multi-output architecture comparison against MLP and CNN-GAF baselines, separating in-sample fit from out-of-sample generalisation.

NourishMama — Cloud ELT Pipeline & Data Warehouse

Python · GCP (GCS, BigQuery) · Terraform · Bruin · Streamlit

- Built a Bruin-orchestrated ELT pipeline on GCP with Terraform, loading 8,000+ records into a layered BigQuery warehouse with data quality tests at each layer, feeding a Streamlit dashboard for non-technical users.

EXPERIENCE

Lecturer — Data Analytics & Machine Learning (MSc)

Oct 2024 – Jul 2026

Berlin School of Business & Innovation, Berlin, Germany

- Designed and delivered graduate curriculum in applied statistics, machine learning and data pipelines; students built end-to-end analytical projects using Python, SQL and cloud tools.
- Mentored students on reproducible workflows, model evaluation, data modelling and communicating quantitative results to non-technical stakeholders.

Senior Research Associate / Visiting Assistant Professor

Aug 2022 – Aug 2023

Illinois Institute of Technology, Chicago, USA

- Developed modular, reusable Python components for numerical simulation workflows in an international research collaboration.
- Taught undergraduate Calculus to classes of 40+ students; designed and graded assessments.

Research Scientist

Dec 2018 – Jul 2022

Johann Radon Institute for Computational and Applied Mathematics (RICAM), Austrian Academy of Sciences, Linz, Austria

- Developed high-performance algorithms in Python and MATLAB for high-dimensional numerical integration (quasi-Monte Carlo methods) in a collaboration across Belgium, Austria and the US, resulting in 5 peer-reviewed publications.
- Teaching assistant for Mathematics for AI I, II & III at Johannes Kepler University Linz, supporting 100+ students per course.

Career break Aug 2023 – Oct 2024: international relocation from the United States to Germany.

EDUCATION

PhD Computational Mathematics — Johannes Kepler University Linz, Austria

2019 – 2022

MSc Financial Engineering (part-time) — WorldQuant University

2024 – ongoing

MSc Mathematics — African University of Science and Technology, Nigeria

2016 – 2017

BSc Industrial Mathematics & Computer Science — Ebonyi State University, Nigeria

2010 – 2014

TECHNICAL SKILLS

Machine learning & statistics: scikit-learn, XGBoost, TensorFlow/Keras, regression (incl. regularised), time-series forecasting (LSTM), clustering, PCA, statsmodels, stationarity testing, SHAP

MLOps: MLflow, Prefect, Evidently, Docker, Kubernetes, GitHub Actions (CI/CD)

LLMs & applied AI: LangChain, RAG and agents, FAISS, Hugging Face, OpenAI GPT-4o

Programming: Python (pandas, NumPy, SciPy), SQL, R, MATLAB

Data engineering & cloud: GCP (BigQuery, GCS), Terraform, dbt, Airflow, Bruin, DuckDB, SQLite

Visualisation: Streamlit, Plotly, Matplotlib, Power BI

Languages: English (fluent), German (A2, actively improving)

AWARDS & CERTIFICATIONS

- First Place, Data Science & Machine Learning Hackathon — MIT Institute for Data, Systems, and Society (2024)
- MLOps Zoomcamp — DataTalksClub (2024); Data Engineering Zoomcamp — DataTalksClub (2026)
- Special Prize, Zonta Award — Johannes Kepler University Linz (2021)

SELECTED PUBLICATIONS

- Digit-by-digit and component-by-component construction of lattice rules for periodic functions with unknown smoothness. *Journal of Complexity*, 2021.
- Construction of good polynomial lattice rules in weighted Walsh spaces by an alternative component-by-component construction. *Mathematics and Computers in Simulation*, 2022.
- Component-by-component digit-by-digit construction of good polynomial lattice rules in weighted Walsh spaces. *Constructive Approximation*, 2022.
- Full list (5 peer-reviewed publications): Google Scholar